PARSHWANATH CHARITABLE TRUST'S
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Data Science

CSE DATA SCIENCE

SmartShelf:AI based book detection and recommendation

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Project Guide
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Outline

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Introduction

Exploring books on a shelf can be difficult when users do not know the genre or description of each book. Searching for this information manually for every book is time-consuming, especially when many books are present.

This project develops a web application that analyzes bookshelf images to detect books and extract their titles and authors. It then retrieves details such as genre and description using external book APIs, allowing users to quickly view information about multiple books in one place and discover new reading options.

Introduction

Objectives:

- To detect multiple books from a books image using YoloV8 and extract the title and author using OCR techniques.
- To match the extracted text with online book databases to retrieve accurate details such as genre and description.
- To present the details of all detected books together, allowing users to quickly view genres and summaries of multiple books from a single shelf.
- To provide users with five random book suggestions based on a genre selected by the user.

Literature Survey of the existing system

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Result
1.	[1] YOLOv8 – Latest Object Detection Architecture	Glenn Jocher et al	2023	Achieved significantly faster detection speed compared to earlier YOLO versions. Capable of detecting very small and overlapping objects effectively. Offers simplified deployment and integration in production systems.	It improves upon earlier YOLO versions by using better backbone networks and anchor-free detection mechanisms for higher accuracy and speed. It uses efficient prediction layers and optimized training techniques for improved bounding box accuracy.	Maintained real-time inference ability on modern GPUs and edge devices. Reduced inference time while retaining strong localization accuracy.
2.	[2] A Survey on Book Recommendation Systems	Priyanka Shahane	2021	Content-based methods recommend books similar to user's interests . Cooperative filtering uses similarity between users to suggest books. Hybrid approaches combine multiple techniques to improve accuracy and reduce individual technique drawbacks.	Reviewed multiple existing book recommendation techniques such as content-based, collaborative filtering, demographic, and hybrid methods	Hybrid techniques generally perform better than single-method systems. Recommendation effectiveness increases when combining techniques

Literature Survey of the existing system

Sr. No.	Title	Author(s)	Year	Outcomes	Methodology	Result
3.	[3] Research on a Method for Recognizing Text on Book Spines in Libraries Based on Improved YOLOv11 and Optimized PaddleOCR	Zheng Li, Bingzhen Guo, and Dengcong Mu	2025	Detection performance improved significantly in dense bookshelf scenes. Hyperparameter tuning increased OCR robustness to curved and complex text shapes. The combined system addressed challenges such as overlapping spines, background noise, and special characters.	Proposed an integrated intelligent library text recognition system combining an improved YOLOv11 model with hyperparameter-optimized PaddleOCR. Optimized PaddleOCR parameters (data augmentation, loss weights, character dictionary, input size) for better text recognition	Improved YOLOv11 model achieved better segmentation accuracy than the baseline. Optimized PaddleOCR reduced Character Error Rate (CER) from 8.6% to 3.2%.

Limitations of existing systems

From the literature review of existing systems, we find that :

- **Lack of Integrated Pipeline**: Most existing systems treat detection, OCR, and recommendation separately.
- **Lack of User Personalization and History Tracking**: Many surveyed systems do not store or use user history to personalize recommendations.
- **Cold-Start Problem in Recommendations:** Most existing recommendation methods (collaborative or content-based) fail to suggest relevant books for new users or new books with no interaction history.

Limitations of existing systems

- **Limited Detection in Real-World Scenes:** Traditional object detection models struggle with dense books, overlapping spines, poor lighting, and real library environments
- **Poor Handling of Semantic Meaning :** Traditional TF-IDF based recommendation approaches don’t capture deep semantic relationships between book topics. This can lead to less accurate or irrelevant suggestions.
- **Cold-Start Problem in Recommendations:** Most existing recommendation methods (collaborative or content-based) fail to suggest relevant books for new users or new books with no interaction history.

Problem statement

Readers often come across books on shelves in libraries, bookstores, or personal collections but may not have the time to search each book manually online to understand what it is about.

This project aims to solve this problem by creating a web application that allows users to upload an image of a bookshelf. The system automatically detects the books, extracts the title and author using OCR, and retrieves relevant information such as genre and description using external APIs. This allows users to quickly view important details about multiple books in one place and also receive additional book recommendations based on selected genres.

System Design

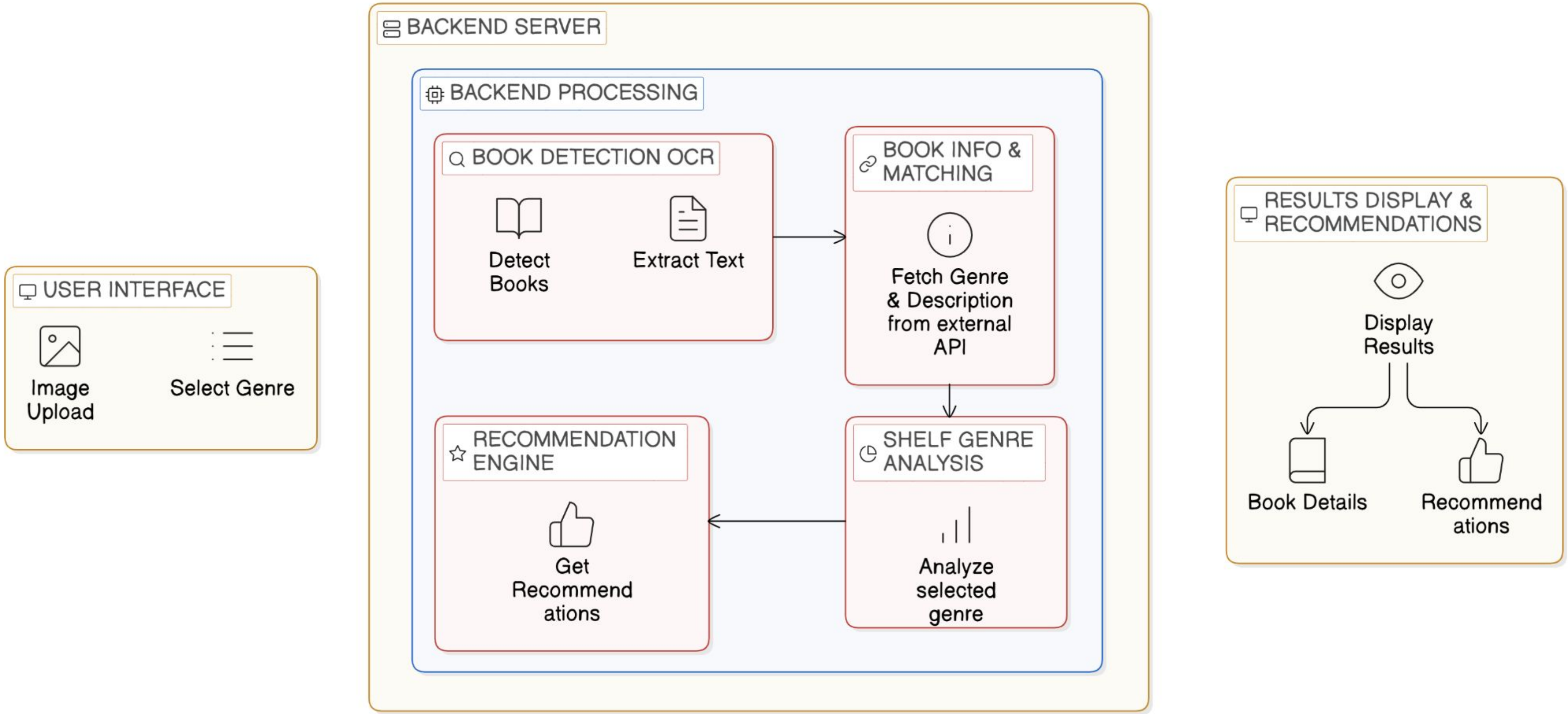


Fig 1: System Design

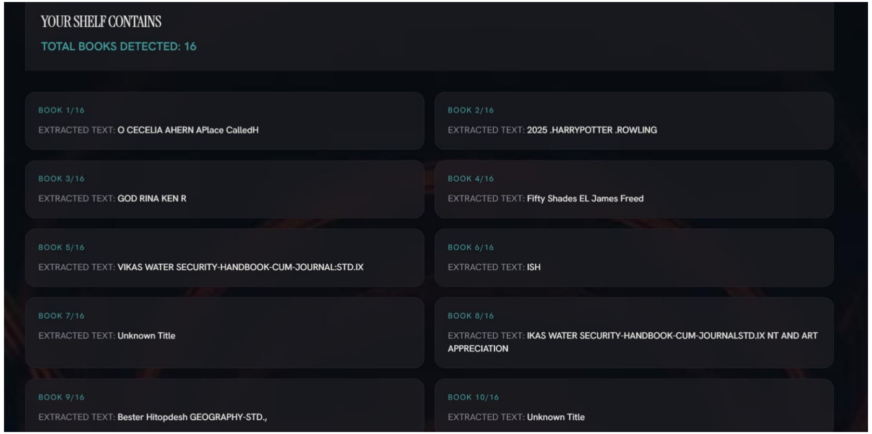
Technologies

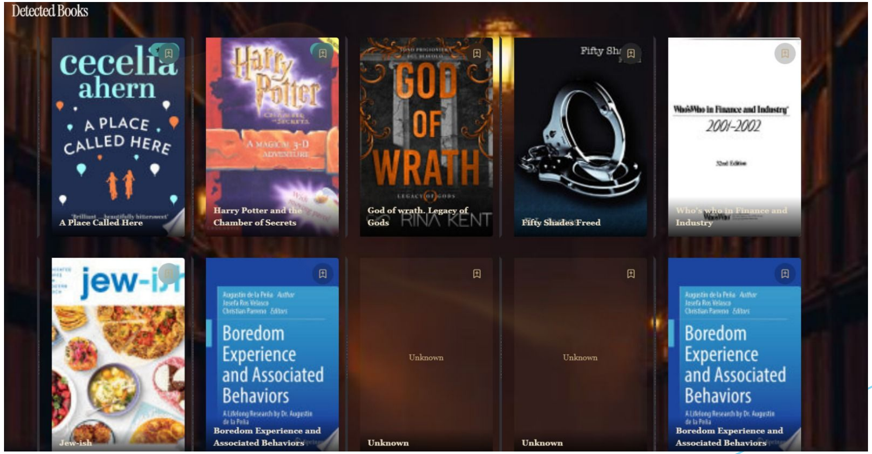
- **Object Detection:** YOLOv8 (Ultralytics)
- **OCR:** PaddleOCR/ EasyOCR
- **Backend Framework:** FastAPI
- **Database:** SQLite
- **Programming Language:** Python 3.10
- **Authentication:** JWT(JSON Web Token)
- **Libraries:** Scikit-learn, Pandas, NumPy

Methodology

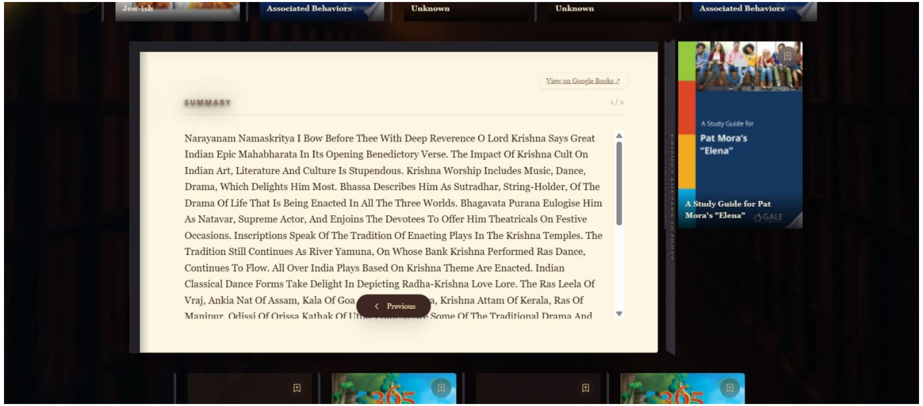
1. **Image Upload :** User upload shelf image.
2. **Object Detection:** YOLO detects books bounding boxes
3. **Text Extraction:** PaddleOCR extracts book title text.
4. **Data Cleaning:** Text preprocessing and normalization.
5. **Feature Engineering:** TF-IDF vectorization for book similarity.
6. **Recommendation Generation:** Cosine similarity used to find similar books.
7. **Database Storage:** Result saved in SQLite tables.

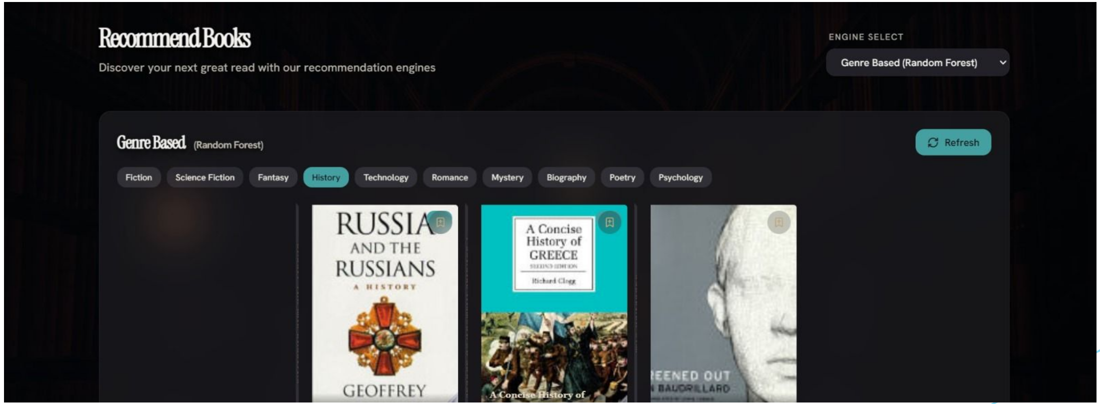
Implementation





Implementation





Conclusion

SmartShelf demonstrates a powerful integration of object detection, OCR, and recommendation systems to create an intelligent and user-friendly solution for book management. By enabling automated book identification from images, it significantly reduces manual effort while improving accuracy and efficiency. The system enhances user experience through personalized book recommendations tailored to individual preferences and maintains a structured history of detections and suggestions for future reference. Its scalable design makes it suitable for diverse applications, including libraries, bookstores, and smart inventory systems, highlighting its potential as a modern, practical solution in the domain of digital book management and recommendation.

References

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Thank You...!!